

Inputs: design vector $\mathbf{x} \in \mathbb{R}^{10}$ | TMY3 weather (.epw) | ASCE 7-22 / ISO loads | RSMears cost coefficients

enter at Tier 1

Tier 1: surrogate search

Path-A

Analytical objectives

- NSGA-III (pymoo)
- ARD Matérn-5/2 GP
- Latin-hypercube init

~338 evaluations

~17 s per case

Pareto front

Tier 2: standards verification

Path-B

Standards-grade Python

- opensee spy FEA
- ISO 13790 hourly EUI
- LM-83 sDA + pplib

789 designs / 3 cases

~0.3 s per design

complete front

Tier 3: high-fidelity check

Path-C

Industry simulator binaries

- EnergyPlus 26.1
- Radiance 6.0 (DC)
- opensee spy P- Δ

789 designs / 3 cases

~12 min per case

verified resource

Outputs: verified Pareto resource (258–272 designs) | three compromise selectors | multi-seed bootstrap HV CIs | cross-tier ρ and IGD